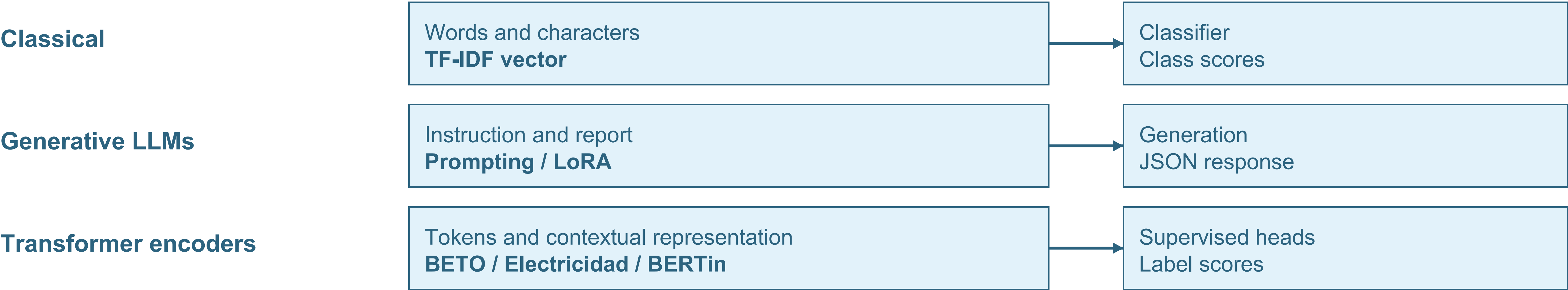


Violence report classification

Models, Severe augmentation and decisions around an anchor

Marcel Herrera Rendón · Universidad de Sonora, México

01 Three families explored



Same 133 reports, joint exact match: TF-IDF+SGD 0.2331; Gemma+LoRA 0.1955; Qwen prompt 0.1053. LoRA pilot: 625 examples, 80 steps. Full S2, macro-F1: character SVM 0.7642 / BETO 0.8198 (threshold 0.45). Different protocols: do not compare panels. [1]

02 Outputs matched to the problem

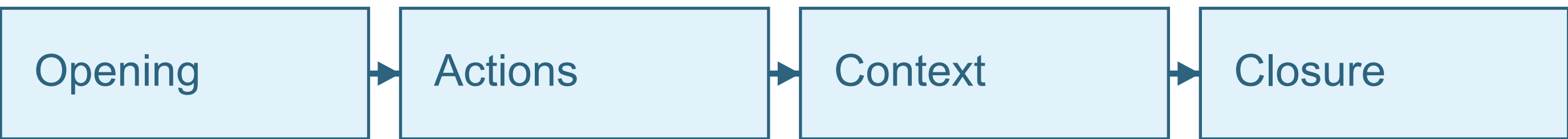
Severity: one ordered category
Mild / Medium / High / Severe
Weighted focal and auxiliary supervision in augrep08

Types: six outputs that can co-occur
Economic, Physical, Property-related, Psychological, Sexual, Vicarious
Final BETO: six sigmoid ≥ 0.45 ; N/A when none is active

Encoders: whitespace normalization, tokenization, truncation and batch padding. Limits: S1 augrep08 512 tokens; final S2 384. [1]

03 Targeted Severe augmentation

616 Severe / 13,630 reports
Training only. Original reports are retained.



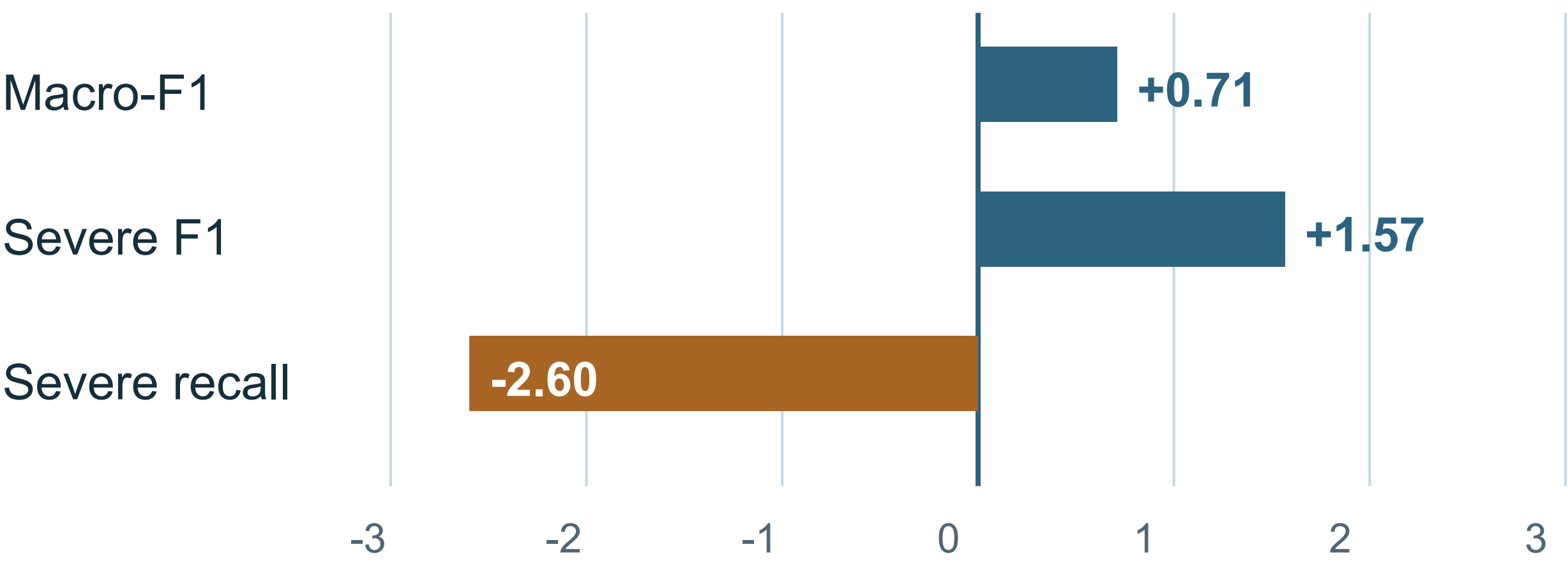
Source cues condition the templates.
Stable seed · At most 2 variants per source
+516 synthetic texts with inherited labels.

4.52% → 8.00% Severe share
1,132 / 14,146 records after augmentation.

Partial rebalancing. These variants use templates, not an LLM. They may introduce facts; label fidelity requires an audit. [2]

Effect on development

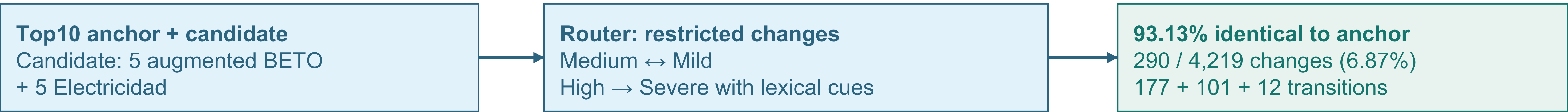
Absolute change × 100 (points)
Same 3,405 reports · Zero = no change



Baseline → augmented: macro-F1 0.5481 → 0.5552;
Severe F1 0.4178 → 0.4335; recall 0.3961 → 0.3701.

F1 rises and recall falls. Style, balance and weights change together.
Controls and repeated seeds must isolate their effects. [2]

04 Selected recipes and an anchor for the final adjustment



Official S1: 0.60678 → 0.60798 macro-F1 (+0.00120). S2 stays unchanged. Final decisions are adjusted around the selected recipes. [3]

Annotator disagreement

72.36% of 1,212 development reports have at least one type with split votes.
Three votes, within-item variance $p(1-p)$. No identities or S1 votes: worker comparisons are unavailable. New analysis after the paper. There are 24 unexplained hard/soft discrepancies. [4]

Complete system · Official test

Severity: 0.60798 macro-F1 · 3rd / 16
Types: 0.87016 micro-F1 · 6th / 15
Test scores informed selection. Cross-institution generalization remains untested. [1]

[1] Herrera Rendón (2026). WomenHelp: methods and results. docs/method.md; docs/results.md.
[2] Augmentation: src/womenhelp_competition/text_augmentation.py; results/evidence.json.
[3] Severity router: scripts/materialize_hailmary_s1_foldbag_router_20260507.py.
[4] New annotation analysis: docs/annotation_consensus.md; scripts/analyze_annotations.py.
github.com/mariron42/clasificador_violencia

